

A Dialogue on AI and Paul Erdős

A philosophical conversation between a modern mathematician with Erdős number 3 and the legendary Paul Erdős on artificial intelligence, autoformalization, and the eternal quest for mathematical beauty.

Setting the Scene

You encounter Erdős in the ethereal space between mathematical worlds, where ideas flow freely across time. The great man sits surrounded by floating equations, his eyes bright with curiosity about the modern mathematical landscape. His characteristic intensity radiates as he contemplates the intersection of human insight and machine computation.

On Autoformalization and The Book

Erdős: “My young friend with Erdős number 3! I’ve been observing your work with these thinking machines. Tell me, what do you make of this autoformalization business? They say these large language models can now translate natural language into Lean code. But I wonder—can they capture the essence of what makes a proof beautiful?”

You: “Paul, it’s remarkable what we can achieve now. We can feed mathematical statements in plain English to an AI system, and it produces formal proofs that systems like Lean or Isabelle can verify. Just this year, systems like Seed-Prover achieved over 50% success on competition mathematics. But I keep thinking about what you said regarding the four color theorem—that computer-assisted proofs aren’t from The Book.”

Erdős: *chuckles and waves his hand dismissively* “Ah yes, The Book! You see, when Appel and Haken used their computer to check all those cases for the four color theorem, they gave us certainty, but no enlightenment. Where was the insight? Where was the *why*? A proof should be like a beautiful sunrise—it should make you see the landscape differently, not just confirm that the sun has indeed risen.”

You: “But Paul, these new AI systems are different. They’re not just doing exhaustive case analysis. Some can now engage in mathematical reasoning, finding novel proofs. There’s even research showing systems that discover new proofs of known results using unexpected approaches. Isn’t finding a new proof, even by machine, potentially from The Book?”

Erdős: *leans forward with interest* “Now that intrigues me! But tell me—when your machine finds this proof, do *you* understand it? Can you see why it’s

beautiful? Or are you simply trusting that the machine has manipulated symbols correctly? The Book contains proofs that illuminate truth, not merely verify it.”

On Understanding versus Verification

You: “That’s exactly the tension I grapple with. In my work with Isabelle/HOL, I spend hours debugging formal proofs, fixing syntax errors, making lemmas work. Sometimes I feel like I’m becoming more of a proof engineer than a mathematician seeking understanding.”

Erdős: *strokes his chin thoughtfully* “This reminds me of the old formalist debates. The symbols must serve insight, not the other way around! Your machines—they manipulate symbols perfectly, but do they *understand* what those symbols mean? When I proved that there are infinitely many primes using Euclid’s elegant construction—taking the product of the first n primes, adding 1, and finding a new prime factor—the beauty wasn’t in the calculation, but in seeing why it must work.”

You: “You’re absolutely right about Euclid’s argument—it’s that brilliant construction, not a diagonal argument. And that’s exactly what worries me about some AI developments. These systems seem to understand sometimes. The new theorem provers can engage in self-correction, learning from mistakes like humans do. They can even participate in mathematical discovery. But you raise the crucial question—is understanding necessary for mathematical truth?”

Erdős: *suddenly animated* “But of course! Mathematical truth without understanding is like having perfect vision but being blind to beauty! When I think about my favorite proof—Euclid’s proof that there are infinitely many primes—I don’t just want to know that it’s true. I want to feel that moment of recognition when the construction reveals why it must be true.”

On AI as Mathematical Collaborator

You: “Yet these AI systems are solving problems we couldn’t solve ourselves. They’ve achieved gold-medal performance on International Mathematical Olympiad problems. Isn’t there value in expanding our reach, even if we don’t fully understand every step?”

Erdős: *pauses, looking conflicted* “You know, I’ve been thinking about this. When I collaborated with my hundreds of coauthors, I didn’t fully understand every detail of their contributions either. Perhaps these machines are just very strange collaborators? Very reliable in computation, completely lacking in intuition.”

You: “That’s a beautiful way to think about it! In autoformalization, I’m not

replacing human insight—I’m augmenting it. The AI helps me translate ideas into formal language, handles the tedious verification steps, catches errors I might miss. But the conceptual work, the creative leaps, the choice of what to formalize and why—that’s still deeply human.”

Erdős: *nods slowly* “Yes, but you must promise me something. Don’t let these machines make you lazy! The danger I see is mathematicians becoming consumers rather than creators of insight. If your AI generates a proof of, say, the Riemann Hypothesis, and you can verify it’s correct, but you can’t explain to a bright graduate student why it works—have we really advanced mathematics?”

On Surprise and Discovery

You: “That’s exactly what worries me about the push toward automated theorem proving, where the goal seems to be churning out more and more formal proofs. But as you always said, it’s not just about proving theorems—it’s about understanding why they’re true.”

Erdős: *brightens* “Exactly! And this is where I think your approach might actually serve The Book. If these machines can help translate between natural and formal language, they could make beautiful proofs more accessible. Imagine—a student could ask their AI tutor to explain a complex proof at just the right level of detail for their understanding.”

You: “I love that vision! Instead of AI replacing mathematical intuition, it could help cultivate it. These systems could serve as bridges between formal rigor and human insight. They could help more people experience those ‘aha!’ moments that make mathematics beautiful.”

Erdős: *stands up excitedly* “Now you’re talking! But tell me—in this brave new world of AI mathematics, what happens to surprise? To serendipity? Some of my best discoveries came from unexpected connections, happy accidents, conversations that took surprising turns.”

You: “That’s fascinating you ask. I’ve noticed that AI systems sometimes make connections I wouldn’t have thought of. They might link results from different fields in surprising ways, suggest lemmas I hadn’t considered. But you’re right to worry—there’s something about the random, playful exploration that characterizes human mathematical curiosity that seems hard to replicate.”

On the Future of Mathematical Collaboration

Erdős: *sits back down, looking thoughtful* “Perhaps the key is in how we frame the collaboration. If we use AI to automate the tedious parts—the symbol manipulation, the case checking, the formal verification—we might free ourselves

to focus on the creative, insightful work. The machines could handle the calculations while we focus on the conjectures.”

You: “Exactly! In my experience, Isabelle doesn’t make discoveries for me—it helps me test my ideas more rigorously. It catches mistakes I might miss, suggests tactics I hadn’t considered, helps me explore the consequences of my hypotheses. It’s like having a very patient, very thorough research partner.”

Erdős: *chuckles* “A partner who never gets tired, never needs coffee, and never disagrees with you about politics! But also one who never has those mysterious flashes of insight, never wakes up at 3 AM with a brilliant idea, never gets excited about a beautiful proof.”

You: “True, but maybe that’s not the right expectation. When you worked with other mathematicians, you didn’t expect them all to be exactly like you. Maybe AI systems are just a new kind of mathematical tool—more sophisticated than pencil and paper, but still fundamentally in service of human mathematical understanding.”

Erdős: *nods approvingly* “I think you’re onto something. The danger would be if we started believing the tool was more important than the mathematics. Or if we became so dependent on formal verification that we lost our ability to recognize truth and beauty without mechanical confirmation.”

On Revolutionary Insights

You: “Yes! And there’s another issue—these AI systems are trained on existing mathematics. They might help us formalize and verify what we already know, but will they help us discover genuinely new mathematical worlds? Or will they just help us explore variations on familiar themes?”

Erdős: *eyes lighting up* “Now that’s the crucial question! Mathematics advances through bold leaps into the unknown. When Cantor conceived of different infinities, when Riemann imagined curved spaces—these weren’t just logical deductions. They were acts of imagination that restructured how we think about mathematical reality.”

You: “Which brings us back to understanding versus verification. A truly revolutionary mathematical insight might initially seem counterintuitive, might resist formalization, might require us to develop entirely new formal frameworks. AI that’s trained to match existing patterns might miss these paradigm shifts.”

Erdős: *stands and begins pacing* “Precisely! This is why I insist that proofs must provide insight, not just logical certainty. The most important mathematical discoveries don’t just extend our knowledge—they transform our understanding. They give us new eyes to see familiar territories.”

On the Primacy of Understanding

You: “So maybe the future lies not in AI replacing human mathematical intuition, but in creating better partnerships. AI could handle more of the routine verification and exploration, freeing human mathematicians to focus on the conceptual breakthroughs that require genuine creativity and insight.”

Erdős: *stops pacing and looks directly at you* “But you must maintain the primacy of understanding! Never accept a proof—whether from human or machine—that doesn’t help you see why the result must be true. The moment mathematics becomes just a game of symbol manipulation, we lose what makes it magnificent.”

You: “I promise, Paul. The goal isn’t to replace the mathematician’s quest for understanding, but to amplify it. These tools should help us ask better questions, explore more possibilities, and ultimately gain deeper insights into mathematical truth.”

Erdős: *smiles warmly* “Then perhaps we’re not so far apart after all. These thinking machines of yours—if they truly serve human understanding rather than replacing it, if they help reveal the structure behind mathematical beauty rather than obscuring it—then they might indeed help us discover more proofs from The Book.”

He pauses, then adds with characteristic wit

“Just promise me one thing—when your AI finally proves the Riemann Hypothesis, make sure you understand it well enough to explain it to the SF himself!”

On the Eternal Quest

You: “Paul, thank you. This conversation reminds me why I fell in love with mathematics in the first place—not for the machinery of proof, but for the moments of genuine understanding when the universe reveals its hidden patterns. AI or no AI, that wonder remains the heart of our work.”

Erdős: *nods contentedly* “Indeed, my young friend. Another roof, another proof—and now perhaps, another algorithm, another insight! The methods may evolve, but the essential human drive to understand the mathematical cosmos—that is eternal.”

The equations around him shimmer and dance, as if reflecting the excitement of mathematical discovery across all possible worlds.

You: “And perhaps that’s the real gift of AI in mathematics—not its computational power, but its potential to remind us what makes human mathematical understanding so precious and irreplaceable.”

Erdős: *with a final twinkle in his eye* “Exactly. The machines can calculate, but only we can wonder. They can verify, but only we can see beauty. They can prove theorems, but only we can glimpse The Book.”

Conclusion

This dialogue explores the tension between computational power and mathematical understanding in our age of artificial intelligence. Through Erdős’s distinctive perspective—his love for elegant proofs, collaborative mathematics, and the eternal quest for insight—we examine how AI tools like autoformalization systems might serve human mathematical understanding rather than replace it.

The conversation reveals that while AI may achieve remarkable computational feats in theorem proving and formal verification, the uniquely human capacity for mathematical insight, aesthetic judgment, and paradigm-shifting creativity remains irreplaceable. In this light, AI becomes not a replacement for human mathematical consciousness, but rather its perfect complement—handling the computational burden while freeing the mind for its highest calling: the search for proofs from The Book.

Dialogue conducted on September 26, 2025